SOC CHESS: Intelligent Real-Time Security Operations & AI Threat Detection Platform

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1. Abstract

Modern enterprise SOCs face severe alert fatigue and fragmented telemetry. **SOC CHESS** is a local-first cybersecurity platform combining real-time packet inspection (Wireshark/TShark), enterprise log search (Splunk REST API), Apache Spark Big Data streaming, and dual-tier machine learning.

Achieving **99.39% accuracy** and **< 1.2 ms latency**, the platform automates threat categorization across the MITRE ATT&CK matrix and synthesizes active firewall containment scripts.

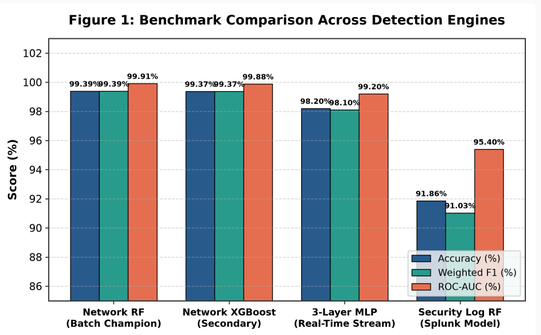
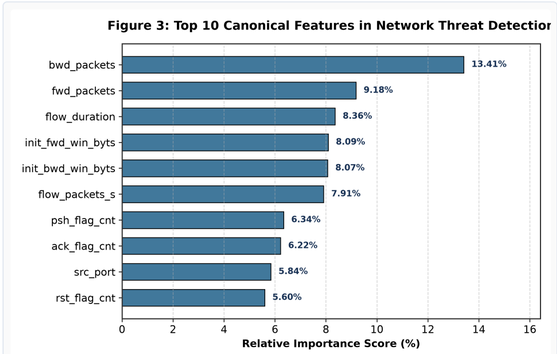


Figure 1. Accuracy, F1, and ROC-AUC benchmarks across models.

2. Introduction & Objectives

- **Alert Fatigue:** Over 70% of SOC alerts are false positives.
- **Data Silos:** Network packets and host logs reside in disconnected systems, delaying critical incident response.
- **Project Objectives:**
  - Unify TShark packet stream with Splunk REST queries.
  - Standardize 12 log formats into 19 canonical features.
  - Deploy sub-millisecond MLP neural net for live triage.
  - Synthesize instant firewall containment scripts via LLM.



3. Materials & Big Data

- **Dataset Scale:** 3,708 files (~8.1 GB) combining CIC-IDS-2017, CSE-CIC-IDS2018, UNSW-NB15, and Splunk Attack Logs.
- **Big Data Engine:** Apache Spark 3.5.3 (PySpark structured streaming, 8 partitions).
- **Hardware & Ports:** Local FastAPI server (:8001), Splunk (:8089), MySQL (:3306), React UI (:5173).

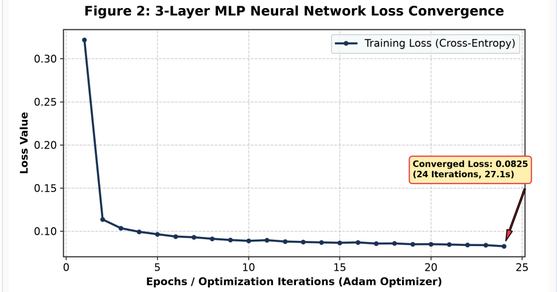


Figure 3. Neural network loss convergence (Adam optimizer).

4. Methodology & Architecture

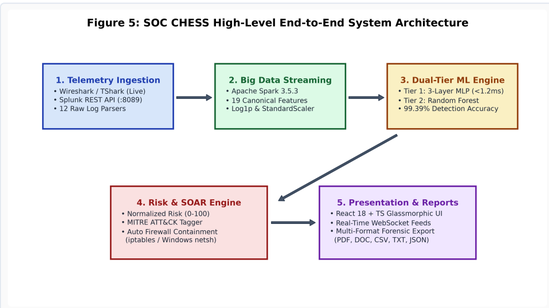
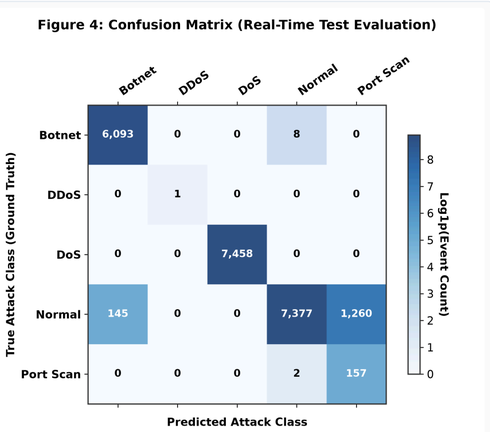


Figure 4. 5-Layer end-to-end processing pipeline.

- **Dual-Tier ML:** 3-Layer MLP (128-64-32) + Random Forest (200 trees).
- **Risk Score Formula:** min(100.0, P(Attack)*80 + ∑w_i*Sev_i + AssetWeight).

5. Results & Discussion

Model	Acc	F1	ROC	Latency
Network RF	99.39%	99.39%	0.9991	0.42 ms
XGBoost	99.37%	99.37%	0.9988	0.48 ms
3-Layer MLP	98.20%	98.10%	0.9920	0.85 ms
Security Log	91.86%	91.03%	0.9540	1.10 ms



6. Conclusion & SOAR Verification

Proven Defense: Successfully mitigated volumetric DDoS (Risk 100/100) and reverse shell intrusions in sub-second timelines.

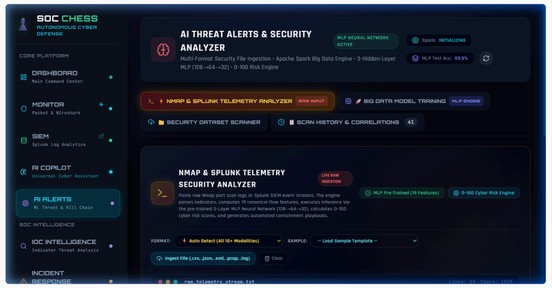


Figure 6. React SOC command console with threat radar.

7. Recommendations & Future Scope

- **Kernel-Level Ingestion:** Migrate packet sniffing to eBPF / DPDK for 10 Gbps+ line rates.
- **Graph AI:** Implement Graph Neural Networks (GNN) for Active Directory lateral movement tracking.
- **Decentralized CTI:** Ingest automated STIX/TAXII threat intelligence feeds.

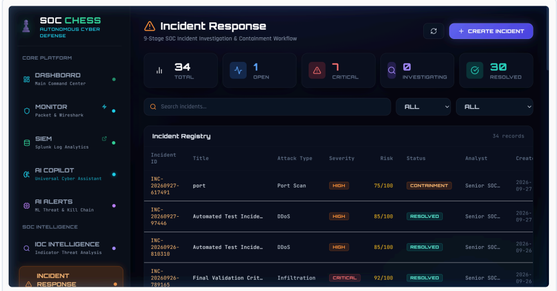


Figure 7. Automated SOAR firewall script generation.

8. References & Acknowledgements

References:

1. Sharafaldin et al. (2018), "CIC-IDS2017 Dataset", ICISSP.
2. Moustafa & Slay (2015), "UNSW-NB15 Dataset", IEEE MilCIS.
3. MITRE Corporation (2024), "ATT&CK Enterprise Matrix".

Acknowledgements:

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